ES 335 / ES 678 · COURSE PROJECT

Project requirements and rubric

Use machine-learning ideas to study a real question from any domain. A careful, well-reasoned project matters more than a headline accuracy score.

Key dates

Proposal	**Thursday, October 15, 2026**	Final project	**Tuesday, November 17, 2026**

The proposal form is open through October 15, 2026 at 11:59 PM IST. Final-project submission will open separately.

Choose your track

ES 335 · Undergraduate

Work individually or with one partner (maximum two students). Build an interactive project website and publish it on GitHub Pages. Submit the live site and source-repository links.

The website should have clear sections for the question, data, preprocessing, baseline and methods, evaluation, results, errors, limitations, conclusion, and reproduction steps. Include at least one useful interaction that helps a visitor understand a result.

GitHub Pages hosts static sites. HTML, CSS, and JavaScript are sufficient; a server-side Python or Node service is not required. If you use a framework, configure it to export a static site.

ES 678 · Graduate

Work individually. Submit a four-page, letter-size conference-style research paper written in LaTeX, plus the source and reproducibility materials requested by the portal.

Include a focused research/depth component connected to the same project. Examples include a derivation, sensitivity or ablation analysis, calibration or uncertainty analysis, a failure-mode audit, or deployment-risk analysis.

Download the ES 678 LaTeX starter package (ZIP: template, style file, and setup notes).

The starter adapts the official NeurIPS 2026 style for this course. It is a course format, not a

conference submission; the course limit is four pages total, including references.

> A deep-learning model is not required. Any method is acceptable when its use is justified for the question and data. A modest or negative result can earn full credit when the reasoning and evaluation are sound.

What the project should demonstrate

1. **Frame the problem.** State the question, what one row represents, the target if there is one, when a prediction would be made, and what information is available then.
2. **Understand the data.** Identify the source, size, important variables, access limits, and data-quality issues relevant to your question. Use focused summaries and plots.
3. **Preprocess carefully.** Explain cleaning and transformations. Fit imputation, scaling, feature selection, and similar preprocessing using training data only.
4. **Choose methods with reasons.** Include a simple baseline. Explain the objective or core idea, assumptions, and why each method is appropriate. More complex is not automatically better.
5. **Evaluate fairly.** Define the split or validation scheme before comparing models. Keep final test data out of model selection and tuning. Choose metrics that fit the task and explain them.
6. **Interpret the evidence.** Discuss results in context, inspect representative errors, describe limitations, and avoid treating prediction as proof of causation.
7. **Make the work reproducible.** Provide code or steps, data access instructions, split details, and enough information to follow the main result.

Statistical tests are not mandatory. Use a test only when it answers a specific question, and explain what it can and cannot establish.

Assessment rubric · 100 project points

Both tracks share 80 points for the quality of the machine-learning reasoning. The remaining 20 points assess the required track deliverable and individual understanding.

Shared ML reasoning · 80 points

CRITERION	POINTS	EVIDENCE LOOKED FOR
PROBLEM FORMULATION AND SCOPE	10	A precise question; target and unit are defined; prediction-time information and the limits of the claim are clear.
DATA UNDERSTANDING	10	Source, row meaning, important variables, access limits, and relevant quality checks are explained.
PREPROCESSING AND STATISTICAL REASONING	10	Transformations and summaries are justified; learned preprocessing avoids test-data leakage.
BASELINE AND METHOD SELECTION	15	A meaningful baseline is compared; objectives, assumptions, and method choices fit the question and data.
EVALUATION DESIGN AND METRICS	20	The split or validation design is appropriate; tuning is separate from final testing; metrics and comparisons are justified.
RESULTS, ERRORS, AND LIMITATIONS	10	Results are interpreted in context; representative errors, uncertainty, and limitations are discussed without overclaiming.
REPRODUCIBILITY AND TECHNICAL INTEGRITY	5	The data, split, preprocessing, and model steps can be followed; data and credentials are handled responsibly.
SHARED SUBTOTAL	**80**	**Accuracy is not a standalone criterion; interpret it against the baseline and task.**

ES 335 · Undergraduate deliverable · 20 points

CRITERION	POINTS	EVIDENCE LOOKED FOR
INTERACTIVE WEBSITE	10	The site is clear, navigable, works when published, and explains the problem, evidence, and conclusions.
USEFUL INTERACTION	5	At least one interaction helps visitors inspect or understand a result rather than serving as decoration.
INDIVIDUAL UNDERSTANDING AND DEFENSE	5	Each student explains their contribution, key choices, evaluation, and a limitation, and reasons about a small change
ES 335 TOTAL	**100**	**80 shared points + 20 undergraduate deliverable points.**

ES 678 · Graduate deliverable · 20 points

CRITERION	POINTS	EVIDENCE LOOKED FOR
LATEX PAPER AND SCIENTIFIC COMMUNICATION	5	A clear, complete paper follows the supplied two-column course format and stays within four pages including references.
GRADUATE DEPTH ANALYSIS	10	A substantive analysis adds insight beyond the main score, such as a derivation, sensitivity/ablation, calibration/uncertainty, failure audit, or deployment-risk analysis.
INDIVIDUAL UNDERSTANDING AND DEFENSE	5	The student explains the derivation or analysis, assumptions, evidence, limitations, and can reason about a small change.
ES 678 TOTAL	**100**	**80 shared points + 20 graduate deliverable points.**

Credit reflects correctness, reasoning, evidence, and relevance. A limited error can still receive substantial credit when the main approach is sound. Unsupported claims, missing evaluation, or test-data leakage weaken the evidence.

Proposal checklist

Submit a concise proposal by October 15. Include:

- Working project title, course code, student name and roll number; for an ES 335 pair, include both students.
- The question and intended prediction or analysis.
- Planned data source, what one row represents, approximate size, and whether you can access it.
- One simple baseline, candidate method or methods, and a preliminary evaluation plan with a suitable metric.
- One likely challenge, such as missing data, imbalance, leakage, limited sample size, or privacy.
- ES 335: the interaction planned for the website. ES 678: the proposed research/depth component.

The title and tool choices are working details, not exclusive claims; overlapping project titles are allowed. Confirmed team members edit one shared registration record until the proposal deadline.

Tools, AI, and responsible sharing

AI tools are permitted for brainstorming, coding, debugging, editing, and visualization. Disclose the tools and kinds of assistance used, check important outputs independently, and cite external code or material. You are responsible for every result and may be asked to explain or defend any part of the work in a viva.

Do not publish passwords, API keys, private student information, or data you are not permitted to share. Use a permitted dataset, a safe sample, or precomputed outputs when needed.